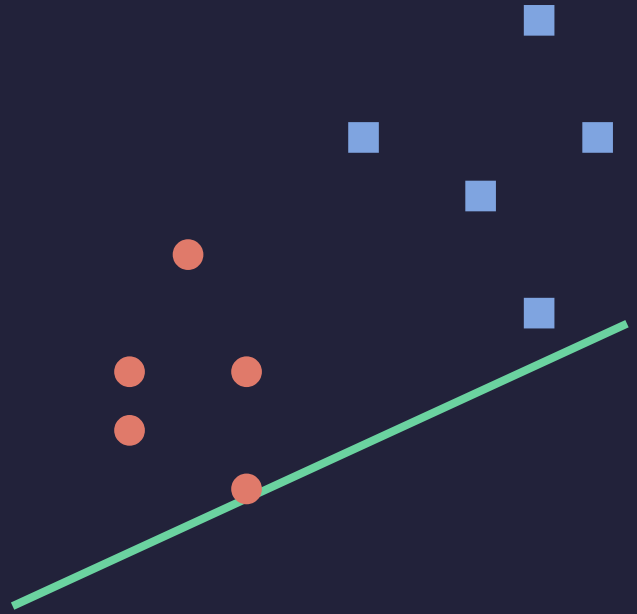
Linear Discriminant Analysis (LDA)



LDA: Introduction

- Linear Discriminant Analysis, also called Normal Discriminant Analysis or Discriminant Function Analysis, is a dimensionality reduction technique that is commonly used for supervised classification problems.
- It is used to project the features in a higher-dimensional space into a lower-dimensional space.

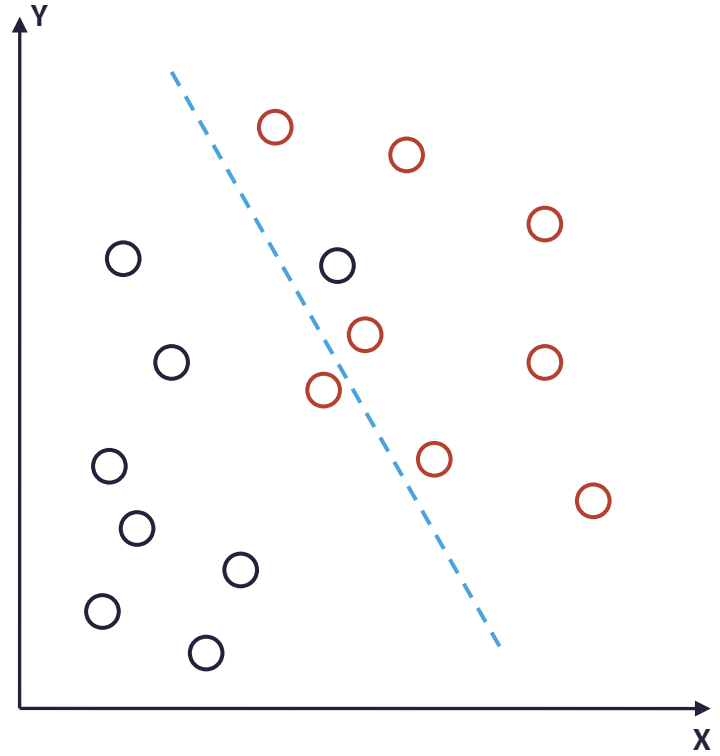
In simple words

- Supervised: LDA uses the class labels of the training data (unlike PCA).
- It finds new axes that best separate the classes.

The Problem

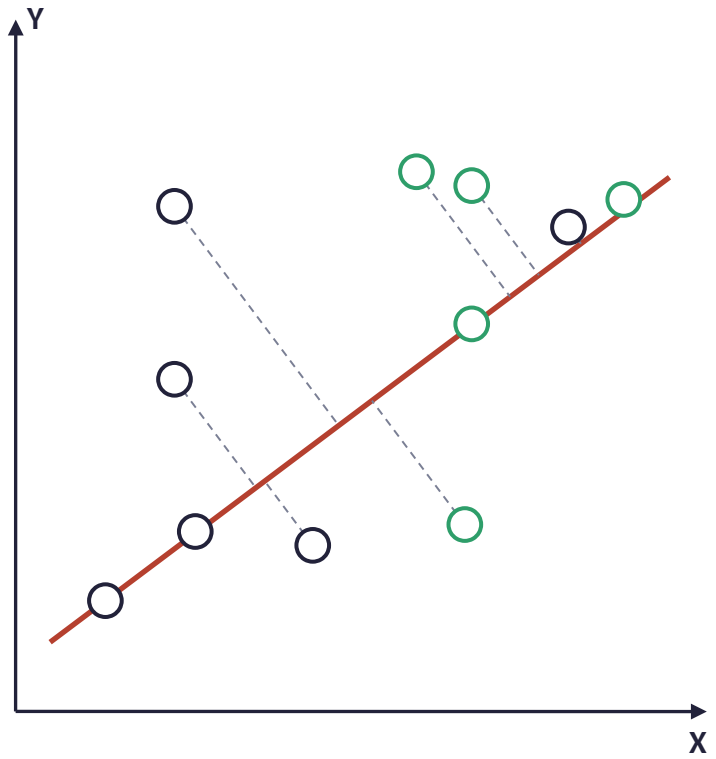
- Suppose we have two sets of data points belonging to two different classes that we want to classify.
- When the data points are plotted on the 2-D plane, there is no straight line that can separate the two classes of the data points completely.

Using only X or only Y, or one straight line in the plane, some points always end up on the wrong side.



Solution: LDA: Creating a New Axis

- Here, Linear Discriminant Analysis uses both axes (X and Y) to create a new axis.
- It projects the data onto the new axis in a way that maximizes the separation of the two categories.
- Hence the 2-D graph is reduced to a 1-D graph.



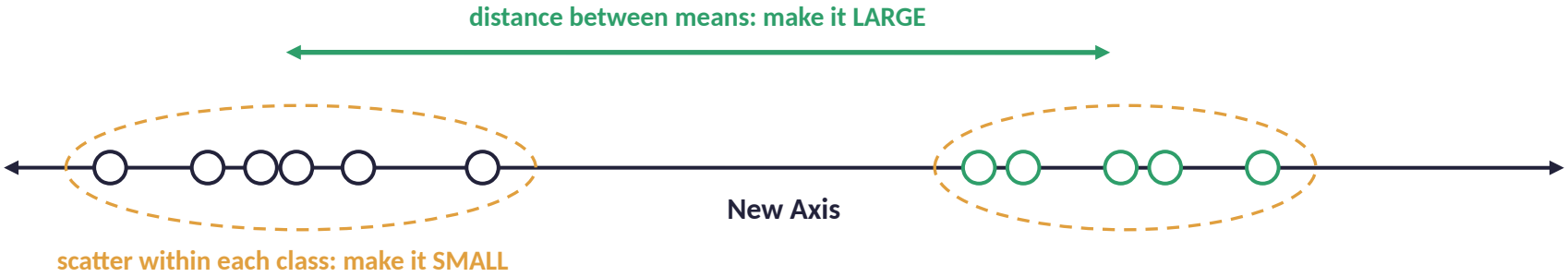
Each point drops onto the red line (dotted lines). Its position along that line is its new single value.

LDA: Two Criteria for the New Axis

Two criteria are used by LDA to create a new axis:

Maximize the distance between the means of the two classes.
→ the two groups are far apart

Minimize the variation (scatter) within each class.
→ each group is tight and compact



LDA: Steps (1–5)

1. Compute the class means of the dependent variable (the mean of each class).

$$\mu_i = \frac{1}{N_i} \sum_{x \in \omega_i} x$$

2. Derive the covariance (scatter) matrix of each class.

$$S_1 = \sum_{x \in \omega_1} (x - \mu_1)(x - \mu_1)^T$$

3. Compute the within-class scatter matrix ($S_1 + S_2$).

$$S_W = S_1 + S_2$$

4. Compute the between-class scatter matrix.

$$S_B = (\mu_1 - \mu_2)(\mu_1 - \mu_2)^T$$

5. Compute the eigenvalues and eigenvectors from the within-class and between-class scatter matrices.

$$S_W^{-1} S_B \, w = \lambda w$$

LDA: Steps (6–8)

6 Sort the eigenvalues and select the top k values.

7 Find the eigenvectors corresponding to the top k eigenvalues.

8 Obtain the LDA by taking the dot product of the eigenvectors and the original data.

Eigenvectors w solve:

$$(S_W^{-1} S_B - \lambda I)\, w = 0$$

λ tells how well that direction separates the classes: the larger λ, the better.

Summary of the idea: Sw measures how spread out each class is; Sb measures how far apart the class means are. The eigenvector of $Sw^{-1}Sb$ with the largest eigenvalue is the direction that gives the biggest between-class distance for the smallest within-class spread.

Solved Example: The Problem

Compute the Linear Discriminant projection for the following two-dimensional dataset.

Class ω_1	x_1	x_2
Sample 1	4	2
Sample 2	2	4
Sample 3	2	3
Sample 4	3	6
Sample 5	4	4

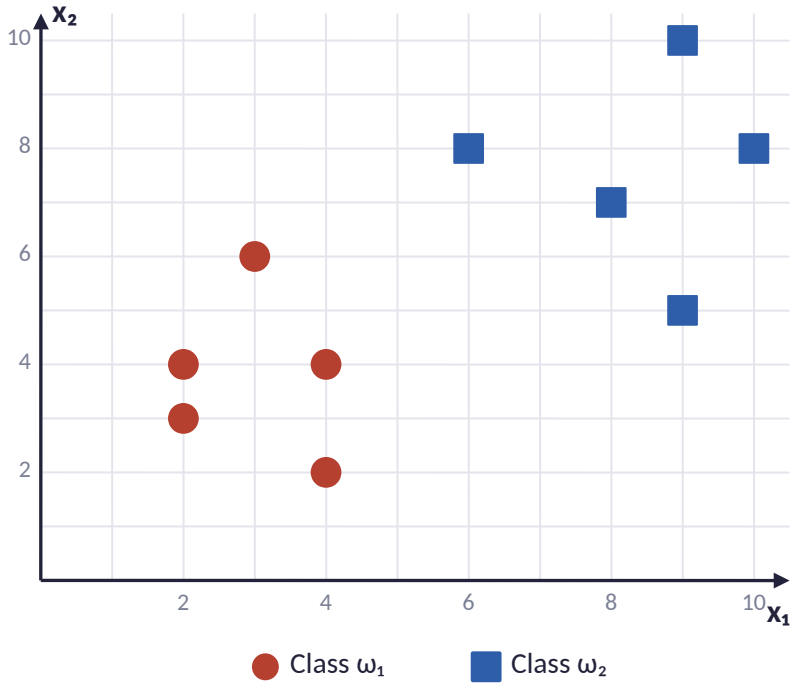
Class ω_2	x_1	x_2
Sample 1	9	10
Sample 2	6	8
Sample 3	9	5
Sample 4	8	7
Sample 5	10	8

- 2 features (x_1, x_2), 2 classes, $N_1 = N_2 = 5$ samples.
- Goal: find the direction w^* and project every sample onto it (2-D $\rightarrow$ 1-D).

In set notation: $X_1 = (x_1, x_2) = \{(4,2), (2,4), (2,3), (3,6), (4,4)\}$ for class ω_1 and $X_2 = (x_1, x_2) = \{(9,10), (6,8), (9,5), (8,7), (10,8)\}$ for class ω_2.

Solved Example: Plotting the Data

Class ω_1: $X_1 = \{(4,2), (2,4), (2,3), (3,6), (4,4)\}$ **Class ω_2:** $X_2 = \{(9,10), (6,8), (9,5), (8,7), (10,8)\}$



- Class ω_1 (red) sits in the lower-left, class ω_2 (blue) in the upper-right.
- The classes overlap in x_2: e.g. (3, 6) in ω_1 is higher than (9, 5) in ω_2.
- So x_1 alone or x_2 alone is not the best single feature – LDA will find a better combined direction.

Step 1: The Class Means

Class ω_1: $X_1 = \{(4,2), (2,4), (2,3), (3,6), (4,4)\}$ **Class ω_2:** $X_2 = \{(9,10), (6,8), (9,5), (8,7), (10,8)\}$

$$\mu_i = \frac{1}{N_i} \sum_{x \in \omega_i} x$$

$$\mu_1 = \frac{1}{5}[(4,2) + (2,4) + (2,3) + (3,6) + (4,4)] = \frac{1}{5}(15,\ 19) = (3,\ 3.8)$$

$$\mu_2 = \frac{1}{5}[(9,10) + (6,8) + (9,5) + (8,7) + (10,8)] = \frac{1}{5}(42,\ 38) = (8.4,\ 7.6)$$

Check: x_1 values of ω_1 add to 4 + 2 + 2 + 3 + 4 = 15 → 15 / 5 = 3; x_2 values add to 2 + 4 + 3 + 6 + 4 = 19 → 19 / 5 = 3.8. So μ_1 = (3, 3.8) and μ_2 = (8.4, 7.6).

Step 2: Covariance Matrix of Class ω_1

Class ω_1: $X_1 = \{(4,2), (2,4), (2,3), (3,6), (4,4)\}$ **Class ω_2:** $X_2 = \{(9,10), (6,8), (9,5), (8,7), (10,8)\}$

$$S_1 = \frac{1}{N_1 - 1} \sum_{x \in \omega_1} (x - \mu_1)(x - \mu_1)^T$$

Divide each sum by N − 1 = 4:

$$S_1 = \begin{bmatrix} 1 & -0.25 \\ -0.25 & 2.2 \end{bmatrix}$$

= (1/4) × [[4.0, −1.0], [−1.0, 8.8]]

Each sample adds $(x - \mu)(x - \mu)^T$, a 2×2 matrix of the products above.

Step 2: Covariance Matrix of Class ω_2

Class ω_1: $X_1 = \{(4,2), (2,4), (2,3), (3,6), (4,4)\}$ **Class ω_2:** $X_2 = \{(9,10), (6,8), (9,5), (8,7), (10,8)\}$

$$S_2 = \frac{1}{N_2 - 1} \sum_{x \in \omega_2} (x - \mu_2)(x - \mu_2)^T$$

Divide each sum by N − 1 = 4:

$$S_2 = \begin{bmatrix} 2.3 & -0.05 \\ -0.05 & 3.3 \end{bmatrix}$$

= (1/4) × [[9.2, −0.2], [−0.2, 13.2]]

Each sample adds $(x - \mu)(x - \mu)^T$, a 2×2 matrix of the products above.

Step 3: Within-Class Scatter Matrix

Class ω_1: $X_1 = \{(4,2), (2,4), (2,3), (3,6), (4,4)\}$ **Class ω_2:** $X_2 = \{(9,10), (6,8), (9,5), (8,7), (10,8)\}$

$$S_W = S_1 + S_2$$

$$Sw = \begin{bmatrix} 1 & -0.25 \\ -0.25 & 2.2 \end{bmatrix} + \begin{bmatrix} 2.3 & -0.05 \\ -0.05 & 3.3 \end{bmatrix} = \begin{bmatrix} 3.3 & -0.3 \\ -0.3 & 5.5 \end{bmatrix}$$

Meaning: Sw adds up the spread of both classes around their own means. LDA wants a direction where this spread, after projection, is as small as possible.

Step 4: Between-Class Scatter Matrix

Class ω_1: $X_1 = \{(4,2), (2,4), (2,3), (3,6), (4,4)\}$ **Class ω_2:** $X_2 = \{(9,10), (6,8), (9,5), (8,7), (10,8)\}$

$$S_B = (\mu_1 - \mu_2)(\mu_1 - \mu_2)^T$$

$$= \begin{bmatrix} 3 - 8.4 \\ 3.8 - 7.6 \end{bmatrix} \begin{bmatrix} 3 - 8.4 & 3.8 - 7.6 \end{bmatrix} = \begin{bmatrix} -5.4 \\ -3.8 \end{bmatrix} \begin{bmatrix} -5.4 & -3.8 \end{bmatrix} = \begin{bmatrix} 29.16 & 20.52 \\ 20.52 & 14.44 \end{bmatrix}$$

How: $(-5.4)^2 = 29.16$; $(-5.4)(-3.8) = 20.52$; $(-3.8)^2 = 14.44$.

Meaning: Sb measures how far apart the two class means are. LDA wants this to be as large as possible after projection.

Step 5: Find the Eigenvalues

$$S_W^{-1} S_B w = \lambda w \qquad |S_W^{-1} S_B - \lambda I| = 0$$

(a) Inverse of Sw:

$$S_W^{-1} = \frac{1}{(3.3)(5.5) - (-0.3)^2} \text{adj}(S_W) = \frac{1}{18.06} \text{adj}(S_W)$$

$$Sw = \begin{bmatrix} 3.3 & -0.3 \\ -0.3 & 5.5 \end{bmatrix} \qquad Sw^{-1} = \begin{bmatrix} 0.3045 & 0.0166 \\ 0.0166 & 0.1827 \end{bmatrix}$$

(b) Multiply Sw^{-1} Sb:

$$\begin{bmatrix} 0.3045 & 0.0166 \\ 0.0166 & 0.1827 \end{bmatrix} \begin{bmatrix} 29.16 & 20.52 \\ 20.52 & 14.44 \end{bmatrix} = \begin{bmatrix} 9.2213 & 6.489 \\ 4.2339 & 2.9794 \end{bmatrix}$$

(c) Solve $|Sw^{-1}Sb - \lambda I| = 0$:

$$\begin{bmatrix} 9.2213 - \lambda & 6.489 \\ 4.2339 & 2.9794 - \lambda \end{bmatrix}$$

$$(9.2213 - \lambda)(2.9794 - \lambda) - 6.489 \times 4.2339 = 0$$

$$\lambda^2 - 12.2007\lambda = 0$$

$$\lambda(\lambda - 12.2007) = 0$$

$\lambda_1 = 0, \ \lambda_2 = 12.2007$

One eigenvalue is 0 because Sb has rank 1 (two classes → one useful direction).

Step 6: Find the Eigenvectors

$$(S_W^{-1} S_B - \lambda I)\, w = 0$$

$$\lambda = 0: \quad 9.2213\, w_1 + 6.489\, w_2 = 0 \;\Rightarrow\; w_2 = -1.4211\, w_1$$

$$\lambda = 12.2007: \quad (9.2213 - 12.2007)\, w_1 + 6.489\, w_2 = 0 \;\Rightarrow\; w_2 = 0.4591\, w_1$$

Normalise each to length 1:

$$w_1 = \begin{bmatrix} -0.5755 \\ 0.8178 \end{bmatrix}$$

for $\lambda_1 = 0$ → discarded

$$w_2 = \begin{bmatrix} 0.9088 \\ 0.4173 \end{bmatrix} = w^*$$

for $\lambda_2 = 12.2007$ → chosen (largest λ)

Step 7: Project the Data (Dot Product)

Obtain the LDA by taking the dot product of the eigenvector and the original data:

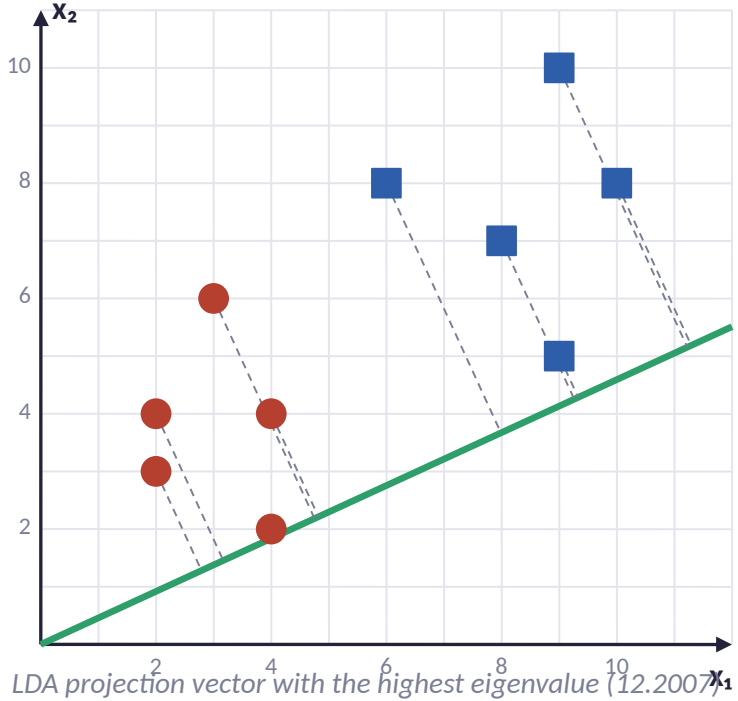
$$y = w^{*T}x = 0.9088\,x_1 + 0.4173\,x_2$$

$[x_1\ x_2]$ (1 × 2) · w^* (2 × 1) = one number (1 × 1)

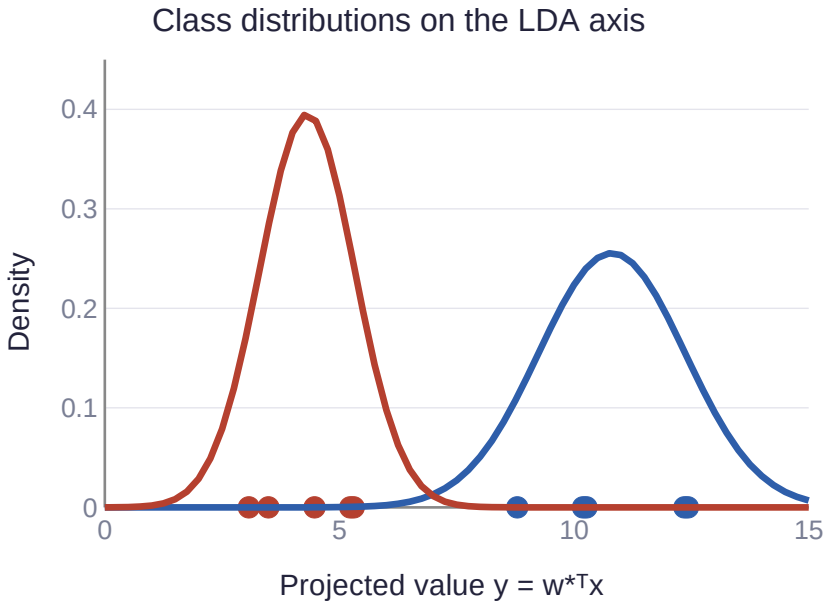
	ω_1	ω_1	ω_1	ω_1	ω_1	ω_2	ω_2	ω_2	ω_2	ω_2
X1	4	2	2	3	4	9	6	9	8	10
X2	2	4	3	6	4	10	8	5	7	8
1st LD	**4.47**	**3.49**	**3.07**	**5.23**	**5.30**	**12.35**	**8.79**	**10.27**	**10.19**	**12.43**

Result: class ω_1 projects to about 3 – 5.3, class ω_2 to about 8.8 – 12.4, so the classes are completely separated on the new 1-D axis.

Visualising the LDA Projection



LDA projection vector with the highest eigenvalue (12.2007)



Dots on the axis are the projected samples; curves are normal fits to each class.

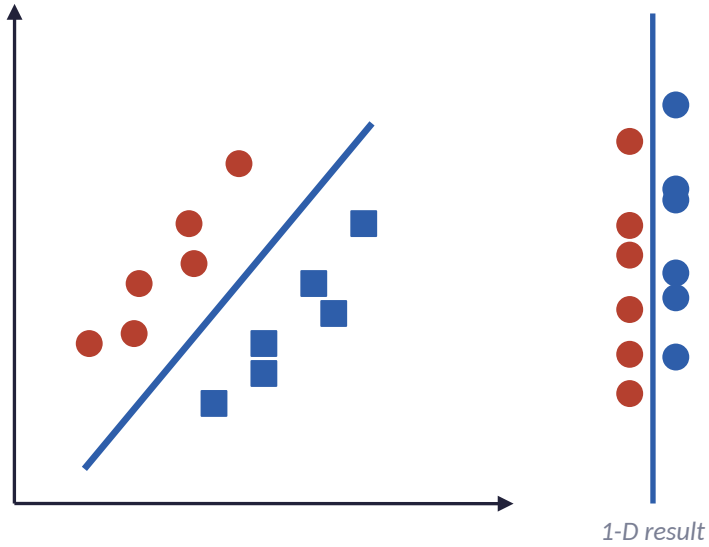
PCA vs LDA

Aspect	PCA	LDA
Type of learning	Unsupervised – ignores class labels	Supervised – uses class labels
Goal	Keep maximum variance in the data	Maximize separation between classes
Finds	Directions of largest spread (principal components)	Directions that best discriminate the classes (linear discriminants)
Matrix used	Covariance matrix S of all data	Sw^{-1} Sb (within- and between-class scatter)
Max. new dimensions	Up to the number of features	At most (number of classes − 1)
Best for	Compression, visualisation, pre-processing, before clustering	Pre-processing for classification; the classifier itself
Weakness	Can lose the direction that separates classes	Needs labels; assumes classes are roughly normal with similar spread

Both are linear methods that reduce dimensions by projecting data onto new axes found with eigenvectors.

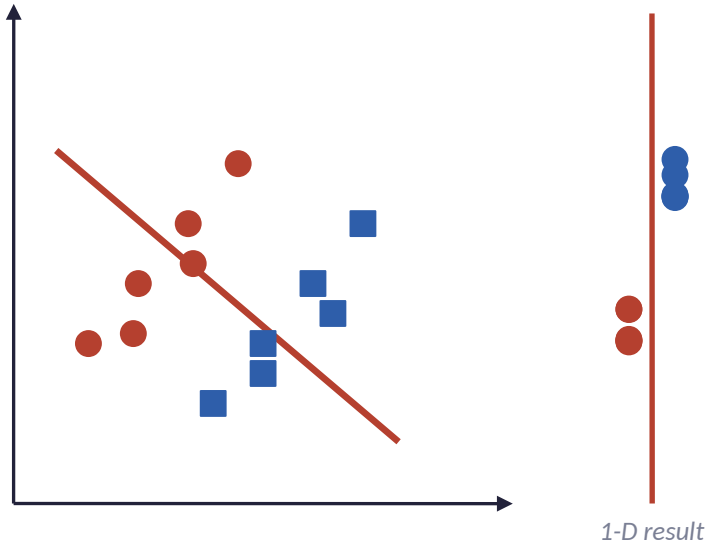
PCA vs LDA on the Same Data

PCA: direction of maximum variance



Follows the long spread of the data – the two classes get mixed after projection.

LDA: direction of maximum separation



Uses the labels – the two classes stay apart after projection.

Key Takeaways

1. LDA is a supervised dimensionality reduction method that uses class labels
2. It finds an axis that maximizes the distance between class means and minimizes within-class scatter
3. $S_w = S_1 + S_2$ (within), $S_b = (\mu_1 - \mu_2)(\mu_1 - \mu_2)^T$ (between); solve $S_w^{-1} S_b w = \lambda w$
4. For 2 classes: $w^* = S_w^{-1}(\mu_1 - \mu_2) = (0.9088, 0.4173)$ in our example
5. PCA keeps variance (unsupervised); LDA keeps class separation (supervised)